

SHETH LUJ AND MV COLLEGE
OS Practical No. 6

Aim:

CPU Scheduling Algorithms (Part 2) – Round Robin

Implement Round Robin scheduling with configurable time quantum.

Compare with FCFS: fairness, turnaround, response time.

Track context switches and improve queue management.

CODE 6A

```
processes = [  
    ["P1", 0, 5],  
    ["P2", 4, 2],  
    ["P3", 5, 4]  
]  
  
quantum = 2  
n = len(processes)  
  
remaining = [5, 2, 4]  
completion = [0, 0, 0]  
waiting = [0, 0, 0]  
turnaround = [0, 0, 0]  
  
queue = []  
time = 0  
completed = 0  
added = [False] * n  
  
gantt = []  
gantt_time = [0]  
  
while completed < n:  
  
    for i in range(n):  
        if processes[i][1] <= time and not added[i]:  
            queue.append(i)  
            added[i] = True  
  
    if not queue:  
        time += 1  
        continue  
  
    i = queue.pop(0)  
  
    start = time
```

```
run = min(quantum, remaining[i])

time += run
remaining[i] -= run

gantt.append(processes[i][0])
gantt_time.append(time)

for j in range(n):
    if processes[j][1] <= time and not added[j]:
        queue.append(j)
        added[j] = True

if remaining[i] > 0:
    queue.append(i)
else:
    completion[i] = time
    completed += 1

for i in range(n):
    turnaround[i] = completion[i] - processes[i][1]
    waiting[i] = turnaround[i] - processes[i][2]

print("\nRound Robin Scheduling")
print("-----")

print("Process\tAT\tBT\tWT\tTAT")

for i in range(n):
    print(f"{processes[i][0]}\t{processes[i][1]}\t{processes[i][2]}\t"
          f"{waiting[i]}\t{turnaround[i]}")

print("\nAverage Waiting Time:", sum(waiting) / n, "ms")
print("Average Turnaround Time:", sum(turnaround) / n, "ms")

print("\nGantt Chart")
print("-----")

for p in gantt:
    print("|", p, end=" ")

print("|")

for t in gantt_time:
```

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```
print(t, end="\t")
```

```
print()
```

OUTPUT

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Round Robin Scheduling

Process	AT	BT	WT	TAT
P1	0	5	2	7
P2	4	2	0	2
P3	5	4	2	6

Average Waiting Time: 1.3333333333333333 ms

Average Turnaround Time: 5.0 ms

Gantt Chart

P1	P1	P2	P1	P3	P3				
0	2	4	6	7	9	11			

CODE 6B

```
print("S091 Shreyash Kadam")
```

```
processes = [
```

```
    ["P1", 0, 5],
```

```
    ["P2", 1, 3],
```

```
    ["P3", 2, 6]
```

```
]
```

```
quantum = 2
```

```
n = len(processes)
```

```
remaining = [5, 3, 6]
```

```
completion = [0, 0, 0]
```

```
waiting = [0, 0, 0]
```

```
turnaround = [0, 0, 0]
```

```
queue = []
```

```
time = 0
```

```
completed = 0
```

Shreyash Kadam

S091

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```
added = [False] * n

gantt = []
gantt_time = [0]

while completed < n:

    for i in range(n):
        if processes[i][1] <= time and not added[i]:
            queue.append(i)
            added[i] = True

    i = queue.pop(0)

    start = time
    run = min(quantum, remaining[i])

    time += run
    remaining[i] -= run

    gantt.append(processes[i][0])
    gantt_time.append(time)

    for j in range(n):
        if processes[j][1] <= time and not added[j]:
            queue.append(j)
            added[j] = True

    if remaining[i] > 0:
        queue.append(i)
    else:
        completion[i] = time
        completed += 1

for i in range(n):
    turnaround[i] = completion[i] - processes[i][1]
    waiting[i] = turnaround[i] - processes[i][2]

print("\nRound Robin Scheduling")
print("-----")

print("Process\tAT\tBT\tWT\tTAT")

for i in range(n):
```

```

print(f'{processes[i][0]}\t{processes[i][1]}\t{processes[i][2]}\t'
      f'{waiting[i]}\t{turnaround[i]}')

print("\nAverage Waiting Time:", sum(waiting) / n, "ms")
print("Average Turnaround Time:", sum(turnaround) / n, "ms")

print("\nGantt Chart")
print("-----")

for p in gantt:
    print("|", p, end=" ")

print("|")

for t in gantt_time:
    print(t, end="\t")

print()

```

OUTPUT

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Round Robin Scheduling

Process	AT	BT	WT	TAT
P1	0	5	7	12
P2	1	3	5	8
P3	2	6	6	12

Average Waiting Time: 6.0 ms

Average Turnaround Time: 10.666666666666666 ms

Gantt Chart

```

-----
| P1 | P2 | P3 | P1 | P2 | P3 | P1 | P3 |
0      2      4      6      8      9      11      12      14
|

```